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Convulsions associated with fever and acute onset of unknown aetiology with case fatalities have become a long observed medical condition at the Child Health Department of the Korle-Bu Teaching Hospital. Children admitted to the department with seizures of undetermined origin and fever has been a source of diagnostic confusion. Studies from the Asia Pacific region suggest a link with non-polio enteroviruses. The aim of the study was to investigate the association between non-polio enterovirus and acute encephalopathy causing neurological morbidity in children. One hundred and fifty cerebrospinal fluid (CSF), throat swab and serum samples were collected from participants at the Child Health Department of the Korle-Bu Teaching Hospital for virus isolation and characterization. Samples were cultured on cells and positive culture assayed by microneutralisation. Direct PCR as well as multiplex PCR were used to detect other viral agents present. Enterovirus isolation rate was approximately 0.67%. Intratypic differentiation by molecular characterization identified a poliovirus from vaccine origin. Further screening by real-time RT-PCR identified the virus as normal Sabin and not vaccine-derive poliovirus. No arbovirus was however detected.

Question: Are human enteroviruses the cause of neurological impairments in children at the Korle-Bu Teaching Hospital?

Answer: Non-polio enteroviruses and chikugunya virus were found not to be the etiologic agent responsible for the convulsion with neurologic morbidity observed in the Ghanaian children. Investigation for other viral agents is recommended.